

Spring 2026A ASTRONOMY 302 Observing Proposal (10 points total)
Add your name to the file name (_LastNameFirstName.docx) then upload to D2L

Write up a short observing proposal using this template describing your group's final project, what science you are trying to achieve, what telescope/instrument are you using, and how much time do you think you need and why? Work with your group members inside and outside of class to come up with a plan, but you must submit a proposal as an individual. Consult your instructors, TAs, or project leads for help with starting your strategy.

Abstract of Scientific Justification (300 words max) (2 points): Summarize your program. Why is this observing program scientifically interesting? → we can know the orbital period and radius (size) of the target planet. What target(s) are you trying to observe and with what observational modes? What science will your program uncover? Distance (how far), velocity, etc.

Telescope Requested (0.5 points): Kuiper

Instrument (0.5 points): Mont4K

Tolerable Moon Brightness (dark, grey, bright, etc) (0.5 points): Write out what moon brightness is acceptable for your program and why. How does the moon impact on the sensitivity of your observations?

41% at 123°

52% at 72°

Target List (2 points)

List potential targets you may observe and their basic properties like brightness. Under visibility write what months/days the target is up. For example “Object can be observed Feb-March.”

Object Name	Right Ascension	Declination	V magnitude	Object Type	Visibility
HD 209458b	22 03 10.7727465312	+18 53 03.549393384	7.7	exoplanet	Oct 3rd
TOI-1516b	22 40 20.2606796232	+69 30 13.455918936	10.9	exoplanet	Oct 2nd

Experimental Design & Technical Description (4.5 points total)

How much time is required for your program and what signal-to-noise is necessary for your observations? (2 points)

The time HD 209458b appears will be 21:22 - 00:26 and TOI-1516b will be 23:42 – 02:28, to increase the quality of the image (make SNR higher), we need to extend the exposure time as much as possible but also short enough ~ 30s (the calculation results we get) and we decide to take ~5s since it's more precise to make sure that we wouldn't miss the dip (the turning point of light curve). Here we find a relative reference star to calculate the flux of the target exoplanet by using $SNR \sim \sqrt{(Flux * Time_exposure)}$. Since its radius should be small to avoid blocking the

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brightness of the exoplanet we are looking for, we take $V_{\text{mag}} = 11$ in the lecture example (it should be taken at the same instrument)

SNR should equal or be greater than 100.

How will these observations contribute toward the accomplishment of the goals outlined in the abstract? (**2 points**)

Back Up Plan. Describe a potential project that can be done with poor weather or archival data. (**0.5 points**)

Process the raw data like classification